

KANGERLUSSUAQ, GreenlandWMO: **042310**Lat: **67.017N**Lon: **50.717W**Elev: **174**StdP: **14.60**Time Zone: **-3.00 (GRL)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-34.5	-29.6	-41.9	0.5	-33.5	-36.8	0.7	-29.3	26.1	27.8	22.4	25.0	8.9	60	0.870

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.9	67.0	50.4	64.8	49.3	62.6	48.5	52.0	63.3	50.5	61.9	49.3	60.3	11.3	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
46.3	46.7	53.1	44.5	43.6	51.7	42.8	40.9	50.9	21.5	63.3	20.6	62.3	19.9	60.4	57.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.6	17.4	15.3	-37.1	70.9	6.7	1.6	-41.9	72.0	-45.8	73.0	-49.6	73.9	-54.4	75.0		
(5)			-37.2	54.4	6.6	1.4	-42.0	55.5	-45.9	56.3	-49.6	57.1	-54.4	58.1	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	24.4	-0.1	-2.4	3.4	21.9	38.9	50.0	51.9	48.1	38.6	23.0	12.5	5.3		
	DBStd	23.12	16.92	17.34	16.13	13.27	8.11	4.75	3.80	4.47	6.54	11.16	14.64	15.63		
	HDD50	9527	1554	1468	1446	844	350	57	24	90	345	838	1124	1387		
	HDD65	14822	2019	1888	1911	1294	809	449	406	525	792	1303	1574	1852		
	CDD50	179	0	0	0	0	6	58	82	30	3	0	0	0		
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0		
	WSAvg	8.3	8.2	7.8	7.1	7.4	8.6	9.4	9.2	8.5	7.8	8.0	8.9	8.7		
	PrecAvg	6.9	0.3	0.2	0.2	0.4	0.5	0.5	1.0	1.3	0.9	0.6	0.5	0.4		
	PrecMax	11.6	0.9	0.9	0.4	1.5	1.3	1.6	4.0	3.1	2.6	2.0	1.5	0.9		
(15) Precipitation	PrecMin	4.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.3	0.2	0.1	0.0	0.0		
	PrecStd	1.8	0.2	0.2	0.1	0.4	0.4	0.5	0.9	0.7	0.6	0.4	0.3	0.2		
	DB	44.2	44.0	46.5	54.7	66.0	69.7	69.6	66.4	59.2	51.2	50.2	44.4			
	MCWB	36.0	36.7	38.2	42.9	48.7	50.4	51.8	50.6	45.5	40.0	40.9	35.2			
	2% DB	37.4	37.8	39.1	47.5	59.5	66.6	67.5	63.8	55.0	46.1	42.8	39.4			
(23) Mean Coincident Wet Bulb Temperatures	MCWB	30.8	30.6	33.0	38.0	45.1	49.9	50.9	49.4	43.6	38.2	35.6	32.9			
	5% DB	31.7	32.5	32.6	43.4	55.5	64.4	65.5	61.2	51.4	41.7	38.4	34.1			
	MCWB	26.4	26.7	27.5	35.9	43.1	48.9	49.9	48.3	42.2	35.2	32.0	29.1			
	10% DB	24.0	24.8	27.3	39.4	51.6	61.4	63.0	58.5	48.5	37.5	34.1	28.4			
	MCWB	20.4	21.2	23.3	33.6	41.2	47.4	49.0	47.3	41.0	32.6	29.3	24.3			
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	36.7	37.3	38.4	43.4	49.3	53.6	54.6	53.5	47.5	41.7	41.3	36.1			
	MCDB	43.1	42.6	46.0	53.4	62.7	64.3	66.4	63.0	55.8	48.6	49.0	42.8			
	2% WB	31.2	30.7	33.1	38.8	46.2	51.3	52.2	50.9	45.3	38.4	35.9	33.1			
	MCDB	37.1	37.4	38.5	45.9	57.6	63.9	63.5	60.4	51.8	45.2	41.9	38.9			
	5% WB	26.1	26.6	27.8	36.5	43.9	49.7	50.9	49.4	43.4	35.5	32.4	29.2			
(33) Mean Daily Temperature Range	MCDB	31.5	32.8	32.7	42.4	53.7	62.4	62.3	58.4	49.4	40.8	37.5	34.4			
	10% WB	20.4	21.2	23.3	34.0	41.8	48.2	49.7	48.0	41.7	32.9	29.2	24.1			
	MCDB	23.9	24.9	27.5	39.0	50.1	59.8	61.1	56.8	47.3	36.7	34.3	28.2			
	5% DB	14.7	15.4	17.2	15.6	15.5	17.4	18.9	17.0	13.5	12.2	13.8	14.3			
	MCDBR	22.9	23.5	21.2	18.2	21.6	23.1	23.6	22.2	17.2	15.7	19.0	19.0			
(40) Clear-Sky Solar Irradiance	MCWBR	18.0	18.7	16.5	11.6	11.6	11.3	11.8	11.6	10.0	10.6	14.2	14.5			
	5% WB	23.1	23.1	21.2	16.9	20.2	21.2	20.6	18.9	14.6	14.2	18.4	18.4			
	MCDBR	18.4	18.5	16.6	11.1	11.5	11.2	11.4	10.8	9.4	10.0	14.1	14.4			
	taub	0.154	0.217	0.246	0.285	0.303	0.314	0.314	0.309	0.271	0.234	0.164	N/A			
	taud	1.919	2.161	2.277	2.250	2.394	2.483	2.534	2.528	2.532	2.260	1.935	N/A			
(42) All-Sky Solar Radiation	Ebn at Noon	77	207	264	278	285	284	281	271	257	196	63	0			
	Edh at Noon	4	15	23	31	31	29	27	24	18	13	3	0			
(44) All-Sky Solar Radiation	RadAvg	18	186	702	1406	1920	2042	1818	1273	711	264	43	0			
	RadStd	1	19	39	89	120	154	161	80	53	25	4	0			

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(46)
		N/A	N/A	N/A	+1.40	N/A	-1.19	N/A	N/A	+40	N/A	
(47)	Regional (2 neighbors)	N/A	N/A	N/A	+1.87	N/A	N/A	N/A	N/A	+40	N/A	(47)

Nomenclature: See separate page